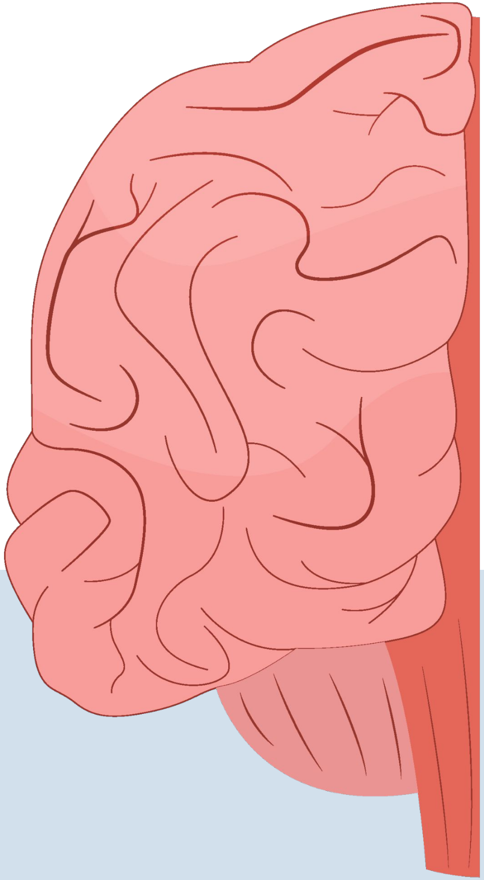


Seizure Detection Using EEG Signals

Kaushika Uppu, Miranda Billawala, Khang Nguyen

CMPE 252 Spring 2025

Introduction

Epilepsy

Neurological disorder characterized by recurring and unprovoked seizures

Seizures can be monitored and detected with electroencephalography (EEG)

EEG

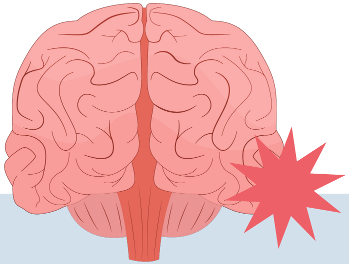
Measures electrical brain activity through electrodes placed on the scalp

EEG signals are very noisy and prone to artifacts (i.e. eye movements)

Objective

Take in raw EEG signals and extract features that describe the data

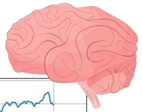
Build a model that can distinguish between normal brain activity and epileptic seizure activity



Overview

Data

CHB-MIT Scalp EEG Database, raw EEG recordings from 22 subjects over multiple days

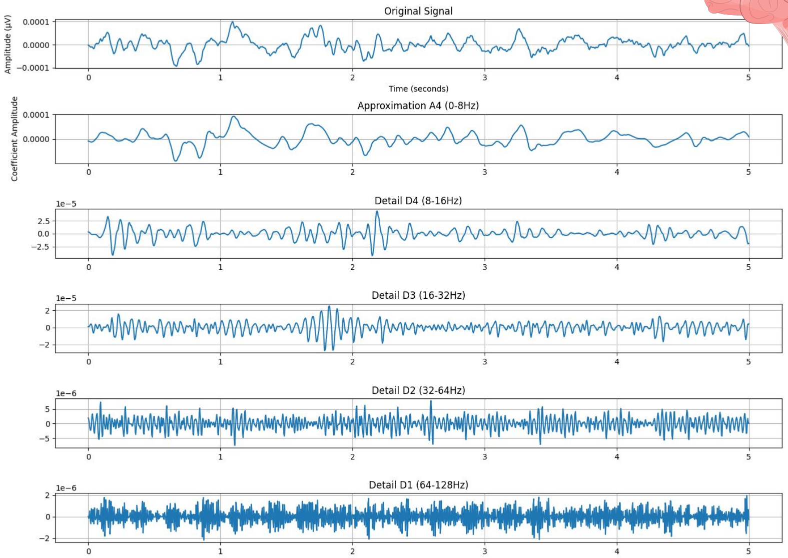


Feature Extraction

Time-domain: patient specific

Frequency-domain: fast Fourier Transforms

4-Level Discrete Wavelet Transform



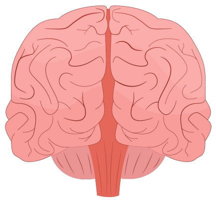
Seizure Detection

Labeled epochs as normal or preictal (period before seizure) using:

(1) Support Vector Machine

(2) XGBoost Classifier

Architecture



Downloading and Importing Raw EEG data

Filtering and Epoching

Artifact Removal (eye blinks, head movements, etc)

Feature Extraction

Initial Model Creation Support Vector Machine and XGBoost

Feature Selection

Hyperparameter Tuning

Final Model

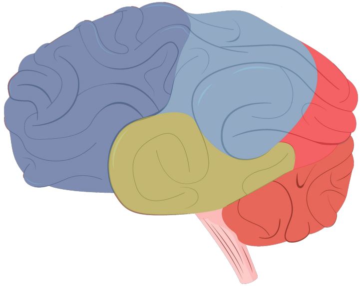
Implementation

Technologies and Tools

- MNE Python library (working with EEG data)
- Support Vector Machines (SVM) for binary classification
- XGBoost Classifier for binary classification

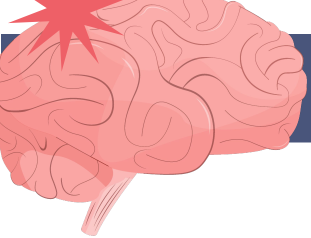
Key Challenges

- Inconsistent EEG channels between patients, standardized to one universal set (resulted in less overall channels)
- Raw EEG data is very large, epoched and saved segments of the data



Design/Technical Decisions

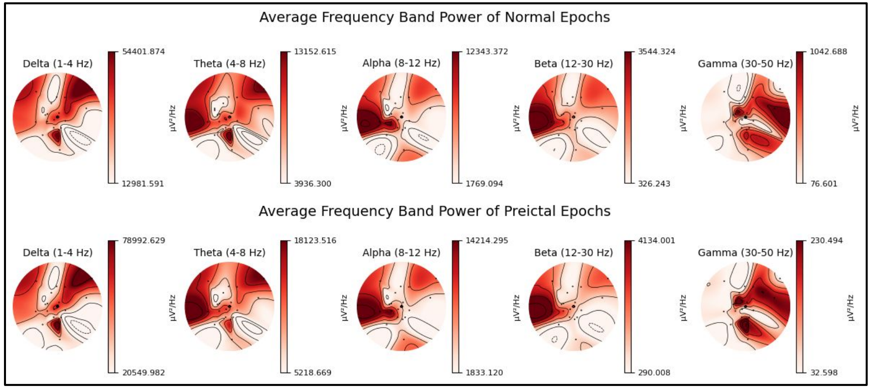
- Training/testing split for each patient (brain activity is very patient-specific)
- Epoching into 5 second intervals, no overlap
- Detecting seizures within 10 minutes of onset



Results

Feature Extraction

One-dimensional, quantifiable data with visualizations to better understand



Model	Metrics				Runtime
	Accuracy	***Precision***	***Recall***	***F1 Score***	
SVM	0.61	0.62	0.84	0.71	2.6s
XGBoost Initial	0.78	0.79	0.84	0.81	19.3s
XGBoost Tuned	0.80	0.80	0.85	0.83	40.4s
XGB and Variance Feature Selection	0.78	0.79	0.83	0.81	8.4s
XGB and 50 KBest Feature Selection	0.70	0.72	0.77	0.75	4.1s
XGB and 100 KBest Feature Selection	0.77	0.78	0.83	0.80	7.5s

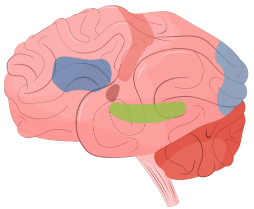
Model Development

SVM - baseline, biased towards preictal

XGBoost - improved performance

Feature Selection - lower performance, faster latency

Conclusion



Key Takeaways

1. Brain signals can be processed to accurately predict a seizure
2. Build specific models to each patient to ensure the best classification

Significance

Manual detection of seizures is very labor and time intensive so having an automated framework makes it more efficient

Future Work

Further feature extraction

Determine average latency of the model; lower model processing time

Create an fully automated pipeline for detecting seizures

Thank You!!